GATE ME 90 Days Challenge

Day 46/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Heat Transfer	**Radiation & Heat Exchangers**	**58 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on black and grey bodies (Stefan–Boltzmann, Wien, Kirchhoff), shape factors (reciprocity, summation, crossed strings), the radiation network, parallel plates and radiation shields, and heat exchangers by the LMTD and ε–NTU methods. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

If the absolute temperature of a black body is doubled, its total emissive power becomes

(A) 2 times
(B) 4 times
(C) 8 times
(D) 16 times

Q2 NAT

A grey surface of emissivity 0.7 is at 600 °C. Taking $\sigma = 5.67 \times 10^{-8}$ W/m²·K⁴, its total emissive power (in kW/m², round off to two decimal places) is _____

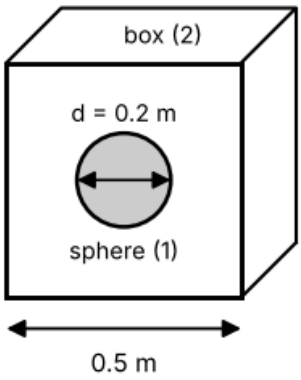
Q3 MCQ

For an opaque grey surface, the correct relation between absorptivity α, reflectivity ρ and emissivity ε is

(A) $\alpha + \rho = 1$ and $\varepsilon = 1 - \rho$
(B) $\alpha + \rho + \varepsilon = 1$
(C) $\alpha = \rho = \varepsilon$
(D) $\varepsilon = 1 - \alpha$

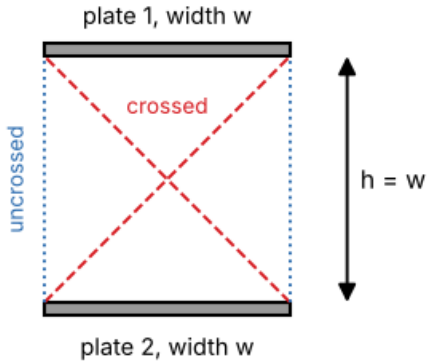
Q4 NAT

A small sphere of 0.2 m diameter is placed inside a closed cubical box of side 0.5 m. The shape factor from the inner surface of the box to the sphere (round off to three decimal places) is _____



Q5 MCQ

Two long parallel plates of equal width w face each other directly at a separation h = w, as shown. By the crossed-strings method, the shape factor between them is



(A) 0.293
(B) 0.414
(C) 0.500
(D) 0.586

Q6 NAT

Two very large parallel plates are at 1000 K (ε = 0.9) and 400 K (ε = 0.5). Taking σ = 5.67×10^{-8} W/m²·K⁴, the net radiation heat flux between them (in kW/m², round off to two decimal places) is _____

Q7 MCQ

The radiation network between two surfaces 1 and 2 of an enclosure consists of, in series,

(A) two space resistances and one surface resistance
(B) $(1-\varepsilon_1)/(\varepsilon_1 A_1)$, $1/(A_1 F_{12})$ and $(1-\varepsilon_2)/(\varepsilon_2 A_2)$
(C) $1/(\varepsilon_1 A_1)$ and $1/(\varepsilon_2 A_2)$ only
(D) $1/(A_1 F_{12})$ only

Q8 NAT

Radiation shields, all with the same emissivity as the two large parallel plates between which they are placed, are used to cut the radiation heat transfer to less than 15 % of its unshielded value. The minimum number of shields needed is _____

Q9 MCQ

In a steam condenser, the steam condenses at constant temperature while cooling water flows through the tubes. For given inlet and outlet temperatures, the LMTD for counter-flow compared with parallel-flow is

(A) larger
(B) smaller
(C) the same
(D) zero

Q10 NAT

In a parallel-flow heat exchanger, the hot fluid cools from 180 °C to 120 °C and the cold fluid is heated from 30 °C to 90 °C. The LMTD (in °C, round off to two decimal places) is _____

Q11 NAT

Oil and water have the same heat capacity rate, 5 kW/K, in a counter-flow heat exchanger. Oil enters at 140 °C; water enters at 40 °C and leaves at 100 °C. With U = 500 W/m²·K, the required heat transfer area (in m²) is _____

Q12 MCQ

The number of transfer units (NTU) of a heat exchanger is defined as

(A) UA/C_{max}
(B) C_{min}/C_{max}
(C) UA/C_{min}
(D) Q/Q_{max}

Q13 NAT

In a parallel-flow heat exchanger, NTU = 1.5 and $C_r = 0.5$, with $C_{min} = 3$ kW/K. The hot fluid enters at 130 °C and the cold fluid at 30 °C. Using $\varepsilon = [1 - \exp(-NTU(1 + C_r))]/(1 + C_r)$, the heat transfer rate (in kW, round off to one decimal place) is _____

Q14 MCQ

The maximum possible heat transfer rate in a heat exchanger is

(A) $C_{max}(T_{h,i} - T_{c,i})$
(B) $C_{min}(T_{h,i} - T_{h,o})$
(C) $C_{min}(T_{h,i} - T_{c,i})$
(D) UA × LMTD

Q15 MCQ

The radiosity of a surface is

(A) the radiation emitted by the surface per unit area
(B) the radiation incident on the surface per unit area
(C) the radiation absorbed by the surface per unit area
(D) the total radiation leaving the surface (emitted plus reflected) per unit area

Answer Key

Q	Type	Answer
Q1	MCQ	(D) 16 times
Q2	NAT	23.07 (accept 22.95 to 23.19)
Q3	MCQ	(A) $\alpha + \rho = 1$ and $\varepsilon = 1 - \rho$
Q4	NAT	0.084 (accept 0.083 to 0.085)
Q5	MCQ	(B) 0.414
Q6	NAT	26.17 (accept 26.04 to 26.30)
Q7	MCQ	(B) $(1 - \varepsilon_1)/(\varepsilon_1 A_1)$, $1/(A_1 F_{12})$ and $(1 - \varepsilon_2)/(\varepsilon_2 A_2)$
Q8	NAT	6
Q9	MCQ	(C) the same
Q10	NAT	74.56 (accept 74.19 to 74.93)
Q11	NAT	15
Q12	MCQ	(C) UA/C_{min}
Q13	NAT	178.9 (accept 177.1 to 180.7)
Q14	MCQ	(C) $C_{min}(T_{h,i} - T_{c,i})$
Q15	MCQ	(D) the total radiation leaving the surface (emitted plus reflected) per unit area

Detailed Solutions

Q1 **Answer: (D) 16 times**

Concept: Stefan–Boltzmann law: $E_b = \sigma T^4$.

$E_b \propto T^4$, so doubling T gives $2^4 =$ **16** times the emissive power.

Wien's law also says the peak shifts to half the wavelength.

Exam tip: Always use kelvin: doubling 300 °C is not doubling the absolute temperature.

Q2 **Answer: 23.07 (accept 22.95 to 23.19)**

Concept: For a grey surface $E = \varepsilon\sigma T^4$ with T in kelvin.

$T = 600 + 273.15 = 873.15$ K; $T^4 = 5.812 \times 10^{11}$ K⁴

$E = 0.7 \times 5.67 \times 10^{-8} \times 5.812 \times 10^{11} = 23\,069$ W/m² = **23.07 kW/m²**

Exam tip: Using T = 873 K instead of 873.15 K changes the answer only in the third significant figure; using °C gives nonsense.

Q3 **Answer: (A) $\alpha + \rho = 1$ and $\varepsilon = 1 - \rho$**

Concept: Radiation falling on a surface is absorbed, reflected or transmitted: $\alpha + \rho + \tau = 1$.

Opaque: $\tau = 0$, so $\alpha + \rho = 1$.

Grey (Kirchhoff): $\varepsilon = \alpha$, hence $\boldsymbol{\varepsilon = 1 - \rho}$.

Exam tip: A polished metal (high ρ) is therefore a poor emitter: that is why radiation shields are polished.

Q4 **Answer: 0.084 (accept 0.083 to 0.085)**

Concept: The sphere (1) is convex and enclosed, so $F_{12} = 1$. Reciprocity: $A_1F_{12} = A_2F_{21}$.

$A_1 = \pi d^2 = \pi(0.2)^2 = 0.1257\ m^2$; $A_2 = 6a^2 = 6(0.5)^2 = 1.5\ m^2$

$F_{21} = A_1/A_2 = 0.1257/1.5 =$ **0.084**

Exam tip: The rest of the box's radiation, $F_{22} = 0.916$, falls back on the box itself.

Q5 **Answer: (B) 0.414**

Concept: Hottel's crossed strings for long surfaces: F_{12} = (sum of crossed strings − sum of uncrossed strings)/(2 × width of surface 1).

Crossed strings (diagonals): $2\sqrt{(w^2 + h^2)} = 2\sqrt{2}\ w$; uncrossed strings (the open sides): $2h = 2w$

$F_{12} = (2\sqrt{2}\ w - 2w)/(2w) = \sqrt{2} - 1 =$ **0.414**

Exam tip: As h/w → 0 the plates become 'infinite' and $F_{12} \to 1$.

Q6 **Answer: 26.17 (accept 26.04 to 26.30)**

Concept: Two-surface enclosure of large parallel plates: $q = \sigma(T_1^4 - T_2^4)/(1/\varepsilon_1 + 1/\varepsilon_2 - 1)$.

$T_1^4 - T_2^4 = 10^{12} - 2.56 \times 10^{10} = 9.744 \times 10^{11}$

$1/0.9 + 1/0.5 - 1 = 2.111$

$q = 5.67 \times 10^{-8} \times 9.744 \times 10^{11}/2.111 =$ **26.17 kW/m²**

Exam tip: The surface with the lower emissivity controls the exchange: raising ε_2 to 0.9 would give 45.2 kW/m².

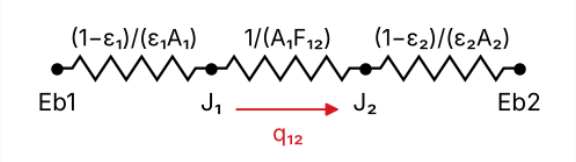
Q7 **Answer: (B) $(1 - \varepsilon_1)/(\varepsilon_1A_1)$, $1/(A_1F_{12})$ and $(1 - \varepsilon_2)/(\varepsilon_2A_2)$**

Concept: Each surface has a surface resistance $(1 - \varepsilon)/(\varepsilon A)$ between E_b and its radiosity J.

Between the two radiosities lies the space (geometric) resistance $1/(A_1F_{12})$.

$q_{12} = (E_{b1} - E_{b2})/[(1 - \varepsilon_1)/(\varepsilon_1A_1) + 1/(A_1F_{12}) + (1 - \varepsilon_2)/(\varepsilon_2A_2)]$.

Exam tip: For a black surface ($\varepsilon = 1$) the surface resistance vanishes and $J = E_b$.



Q8 **Answer: 6**

Concept: With n shields of the same emissivity, there are n + 1 equal resistances in series, so $q_n/q_0 = 1/(n + 1)$.

Need $1/(n + 1) < 0.15 \to n + 1 > 6.67 \to n > 5.67$

Minimum n = **6** (q falls to 1/7 = 14.3 %)

Exam tip: Five shields give 1/6 = 16.7 %, just short: check the boundary case.

Q9 **Answer: (C) the same**

Concept: When one fluid stays at a constant temperature, the terminal temperature differences ΔT_1 and ΔT_2 are the same whichever way the other fluid flows.

Hence the LMTD and the required area are **the same** for counter-flow and parallel-flow.

Likewise $C_r = 0$ and $\varepsilon = 1 - e^{-NTU}$ for every arrangement.

Exam tip: Without phase change, counter-flow always gives the larger LMTD.

Q10 **Answer: 74.56 (accept 74.19 to 74.93)**

Concept: LMTD = $(\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$, with the ΔT at each end of the exchanger.

Parallel flow: inlet end $\Delta T_1 = 180 - 30 = 150$ °C; outlet end $\Delta T_2 = 120 - 90 = 30$ °C

LMTD = $120/\ln 5 = 120/1.6094 =$ **74.56 °C**

Exam tip: In counter-flow the same terminal temperatures would give $\Delta T = 90$ °C at both ends, so LMTD = 90 °C.

Q11 **Answer: 15**

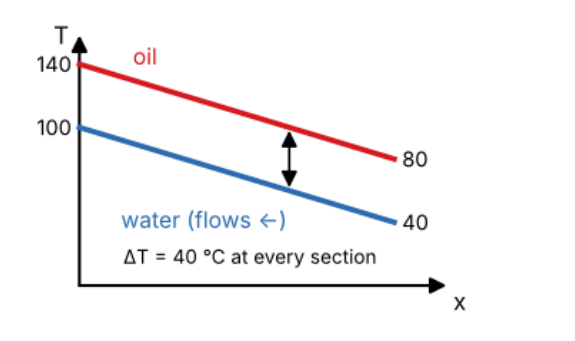
Concept: For balanced counter-flow ($C_h = C_c$), the temperature difference is the same all along the exchanger, so LMTD = ΔT.

$Q = 5 \times (100 - 40) = 300$ kW; oil leaves at $140 - 60 = 80$ °C

$\Delta T = 140 - 100 = 80 - 40 = 40$ °C

$A = Q/(U\Delta T) = 300\,000/(500 \times 40) =$ **15 m²**

Exam tip: The LMTD formula gives 0/0 here; recognise the parallel lines instead.



Q12 **Answer: (C) UA/C_{min}**

Concept: NTU = UA/C_{min}, a measure of the exchanger's thermal size.

Option (B) is the capacity ratio C_r; option (D) is the effectiveness ε.

$\varepsilon = f(NTU, C_r,$ flow arrangement).

Exam tip: Use the ε–NTU method when outlet temperatures are unknown; LMTD then needs trial and error.

Q13 **Answer: 178.9 (accept 177.1 to 180.7)**

Concept: $Q = \varepsilon Q_{max}$, with $Q_{max} = C_{min}(T_{h,i} - T_{c,i})$.

$NTU(1 + C_r) = 2.25$; $e^{-2.25} = 0.1054$

$\varepsilon = (1 - 0.1054)/1.5 = 0.5964$

$Q = 0.5964 \times 3 \times 100 =$ **178.9 kW**

Exam tip: A parallel-flow exchanger can never exceed $\varepsilon = 1/(1 + C_r) = 0.667$ here, however large it is.

Q14 **Answer: (C) $C_{min}(T_{h,i} - T_{c,i})$**

Concept: The largest temperature change any fluid can undergo is $T_{h,i} - T_{c,i}$.

Only the fluid with the smaller heat capacity rate can actually achieve it (in an infinitely long counter-flow exchanger).

So $Q_{max} =$ **$C_{min}(T_{h,i} - T_{c,i})$**; using C_{max} would violate the energy balance of the other fluid.

Exam tip: $\varepsilon = Q/Q_{max}$ lies between 0 and 1 by this definition.

Q15 **Answer: (D) the total radiation leaving the surface (emitted plus reflected) per unit area**

Concept: Radiosity $J = \varepsilon E_b + \rho G$, where G is the irradiation (incident radiation).

It is everything leaving the surface: emitted *and* reflected.

Option (B) is the irradiation G.

Exam tip: Net radiation from an opaque surface $= A(J - G) = (E_b - J)/[(1 - \varepsilon)/(\varepsilon A)]$.

What next?

Day 46 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 47/90: Applied Thermodynamics, Power Cycles. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.